Introduction for the casual reader

A defining step in understanding any complex structure, whether in mathematics or in nature, is to identify the simplest components which do not split any further. Discovering the primary colors, musical notes, or the elementary particles are all different expressions of this same instinct. Then comes our imagination of possible compositions, and the question becomes: are there any rules governing how these simple components can be meaningfully assembled?

Consider the following mathematical theorem: given a spherical object with hair growing from every point on its surface, it is impossible to comb the hair flat without at least one spot left uncombed. For instance, if strands of hair on a globe represent the direction of wind on Earth at any given location, then the theorem guarantees that at any given time there is at least one spot somewhere on Earth where the air is still. Crucially, if the surface of Earth was instead the shape of a doughnut, then it *would* be possible for rotating air flow to cover the entire surface. The wind direction here is a basic example of a *line bundle* on a surface—also known as a vector field in calculus courses. Line bundles are simple structures assembled into higher-dimensional structures called *vector bundles*. This example demonstrates how to learn *homological* information (e.g. the number of holes) about the underlying geometric object (in this case the surface of a planet) through studying the rules governing which vector bundles may be assembled on it.

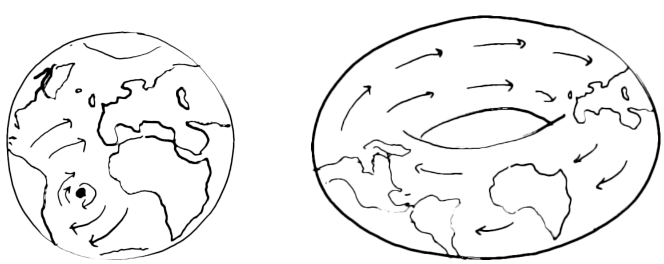

Figure 1: Unlike our planet, cyclones on a toroidal Earth may not have an eye of the storm.

In this thesis, the underlying geometric objects are sets of solutions to systems of polynomial equations which we call *varieties*. For example, the variety defined by the solutions of the polynomial equation $x^2 + y^2 + z^2 = 1$ is a sphere with radius one. A *toric variety* is a special type of variety which enjoys certain kinds of symmetry and arises in applications both within mathematics and across sciences whenever interesting properties are independent of scaling. An early example of this understanding dates back to Renaissance artists and architects who used perspective in their paintings and sketches, essentially discovering the *projective 3-space*, denoted $\mathbb{P}^3$.

From an algebraic perspective, vector bundles on toric varieties are special in that they are also described by systems of polynomial functions. In case of the projective 3-space $\mathbb{P}^3$, these functions are *homogeneous* polynomials in 4 variables with a *standard grading*, such as $wx + yz$ and $x^2y + xyz + yz^2$. This means that all variables have degree 1 and all added terms have the same total degree. The projective n-space $\mathbb{P}^n$ generalizes this space to n-dimensions.

For more general toric varieties, the variables in the polynomials may be *multigraded*: for a toric variety known as the *Hirzebruch surface of type 1*, we have $\text{degree}(x) = \text{degree}(y) = (1, 0)$ while $\text{degree}(z) = (0, 1)$ and $\text{degree}(w) = (-1, 1)$, so that the polynomial $x^2w + yz$ will be homogeneous of degree $(1, 1)$. This introduces a wide range of new possibilities for geometric structures.

The results in this thesis are aimed at extending the classical, standard graded theory over $\mathbb{P}^n$ to the multigraded world of toric varieties by finding the rules for building vector bundles on them. Are there critera for checking whether a vector bundle on a toric variety can be split as a *direct sum* of line bundles? See Chapters 4 and 5. Can a computer find the componentes, or *direct summands*, of the vector bundle? See Chapter 8. Most importantly, am I able to explain an application of my thesis to a non-mathematician?

Consider a second mathematical theorem from linear algebra: an $n \times n$ matrix A is *diagonalizable* (i.e. can be written as $A = PDP^{-1}$ with D a diagonal matrix) if and only if it has n linearly independent eigenvectors. For example, the following is a diagonalization of a 5×5 matrix:

$$\begin{pmatrix} 2 & -3 & 3 & -3 & 3 \\ -3 & 8 & -3 & 10 & 0 \\ 7 & 3 & 2 & -1 & -3 \\ 1 & -3 & 3 & -2 & 3 \\ 13 & -5 & 5 & -13 & -2 \end{pmatrix} = \begin{pmatrix} -3 & -3 & -3 & 0 & 0 \\ 4+2i & 4-2i & 3 & -1 & 1 \\ 3 & 3 & 3 & 1 & 1 \\ -3 & -3 & -3 & 1 & 0 \\ -2-i & -2+i & 1 & -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 2+3i & 0 & 0 & 0 & 0 \\ 0 & 2-3i & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix} \cdot P^{-1}$$

Finding diagonalizations of numerical matrices has applications in almost any field where matrices are used, particularly physics and computer science. What if we instead considered matrices of homogeneous polynomials? The following matrix, for example, represents the system of polynomial functions describing a particular vector bundle on a surface[1]. What does it mean to diagonalize it?

$$A = \begin{pmatrix}
0 & 0 & 0 & 0 & 0 & x_1y_0y_1 & 0 & 0 & x_1y_0y_2 & 0 & 0 & 0 & x_0y_1y_2 & 0 & 0 \\
-x_0y_1+x_1y_1-x_1y_2 & -x_0y_2 & 0 & 0 & 0 & x_1y_2 & x_1y_2 & x_1y_2 & -x_1y_2 & -x_0y_1+x_1y_1-x_1y_2 & -x_1y_2 & 0 & x_1y_2 & x_1y_2 & x_1y_2 \\
0 & 0 & 0 & x_1y_1 & 0 & 0 & 0 & 0 & -x_1y_2 & 0 & -x_1y_2 & 0 & -x_0y_1-x_1y_2 & -x_0y_1 & -x_1y_2 \\
0 & 0 & -x_1y_2 & 0 & x_0y_0-x_1y_0-x_0y_2 & x_1y_2 & x_1y_2 & -x_1y_2 & -x_1y_2 & 0 & x_0y_2-x_1y_2 & x_0y_2 & -x_0y_2+x_1y_2 & x_0y_2+x_1y_2 & 0 \\
0 & 0 & 0 & 0 & 0 & -x_1y_2 & -x_1y_2 & x_1y_2 & x_1y_2 & x_0y_1-x_1y_1+x_1y_2 & x_1y_2 & 0 & -x_1y_2 & -x_1y_2 & -x_1y_2 \\
0 & 0 & x_1y_1 & 0 & x_1y_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & x_0y_1 & x_1y_2 \\
0 & 0 & 0 & 0 & 0 & -x_1y_0 & 0 & -x_1y_0 & -x_1y_0 & 0 & x_0y_2 & x_0y_0 & 0 & 0 & 0 \\
x_0y_1 & x_0y_0 & 0 & 0 & 0 & 0 & 0 & x_1y_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -x_1 & 0 & -x_1 & -x_1 & -x_1 & -x_1 & 0 & x_0-x_1 & x_0 & x_0 & -x_0-x_1 & 0 \\
y_1y_2 & y_0y_2 & 0 & 0 & 0 & -y_0y_1+y_1y_2 & -y_0y_1+y_1y_2 & 0 & 0 & -y_1y_2 & 0 & 0 & -y_1y_2 & y_1y_2 & 0 \\
0 & 0 & y_1y_2 & 0 & y_0y_2 & 0 & y_0y_1 & 0 & y_0y_2 & -y_1y_2 & y_0y_2 & -y_0y_2 & -y_0y_2 & -y_0y_2 & -y_0y_2 \\
-y_1 & -y_0 & 0 & 0 & 0 & 0 & 0 & 0 & y_0-y_2 & y_1 & y_0-y_2 & 0 & -y_0-y_2 & -y_0 & -y_0+y_2 \\
0 & 0 & 0 & -y_1 & 0 & -y_1 & y_1 & 0 & y_2 & 0 & 0 & 0 & y_1+y_2 & y_1 & -y_2 \\
0 & 0 & -y_1 & y_1 & -y_0 & y_1 & y_1 & 0 & y_2 & 0 & y_2 & 0 & -y_1+y_2 & y_1 & -y_2 \\
0 & 0 & 0 & 0 & 0 & -y_0+y_2 & -y_0+y_2 & y_0-y_2 & y_0-y_2 & -y_2 & -y_2 & -y_0 & y_2 & y_2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -y_1 & -y_0 & -y_0 & y_0 & y_0 & y_0-y_2
\end{pmatrix}$$

An application of this thesis is an algorithm for finding invertible matrices P, Q with numerical entries such that $A = PBQ$ and B is the following matrix with blocks along the diagonal:

$$B = \begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
x_1y_2 & x_0y_1-x_1y_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-y_0+y_2 & y_0-y_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & x_0y_1 & x_1y_1-x_1y_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & y_0 & y_0-y_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -x_0+x_1 & x_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & y_0y_2-y_1y_2 & y_0y_1-y_1y_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & x_1y_0 & x_0y_0-x_0y_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -y_1 & -y_1+y_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & x_0y_1-x_1y_1+x_1y_2 & x_0y_2 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -y_1 & -y_0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & x_1y_2 & 0 & x_0y_2 & 0 & x_0y_0-x_1y_0-x_0y_2 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & x_1y_1 & 0 & x_0y_1 & 0 & -x_1y_0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & x_1y_0 & -x_0y_2 & x_0y_0-x_0y_2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_1y_2 & y_0y_1 & 0 & y_0y_1-y_0y_2 & -y_0y_2 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_1 & y_1 & y_1-y_2 & y_1-y_2 & -y_0
\end{pmatrix}$$

Mathematically, this block diagonalization is equivalent to splitting the vector bundle into its simplest components. This example demonstrates that while six of the seven[2] blocks correspond to line bundles, the rightmost block is an *indecomposable* vector bundle on the surface. Learning what governs the existence of such vector bundles, while harder to visualize with a cartoon, is important for studying the geometry of underlying varieties, and a natural motivation for my work.

[1] This is not an introduction for the experts, but if you are one: the sheafification of the cokernel of A over $\mathbb{Z}/3$ is the Frobenius pushforward of the structure sheaf of the blow-up of $\mathbb{P}^2$ in four points as a variety embedded in $\mathbb{P}^1 \times \mathbb{P}^2$.

[2] For those who only count six blocks: if you look carefully, there is a 0×1 block hiding in the top left corner!